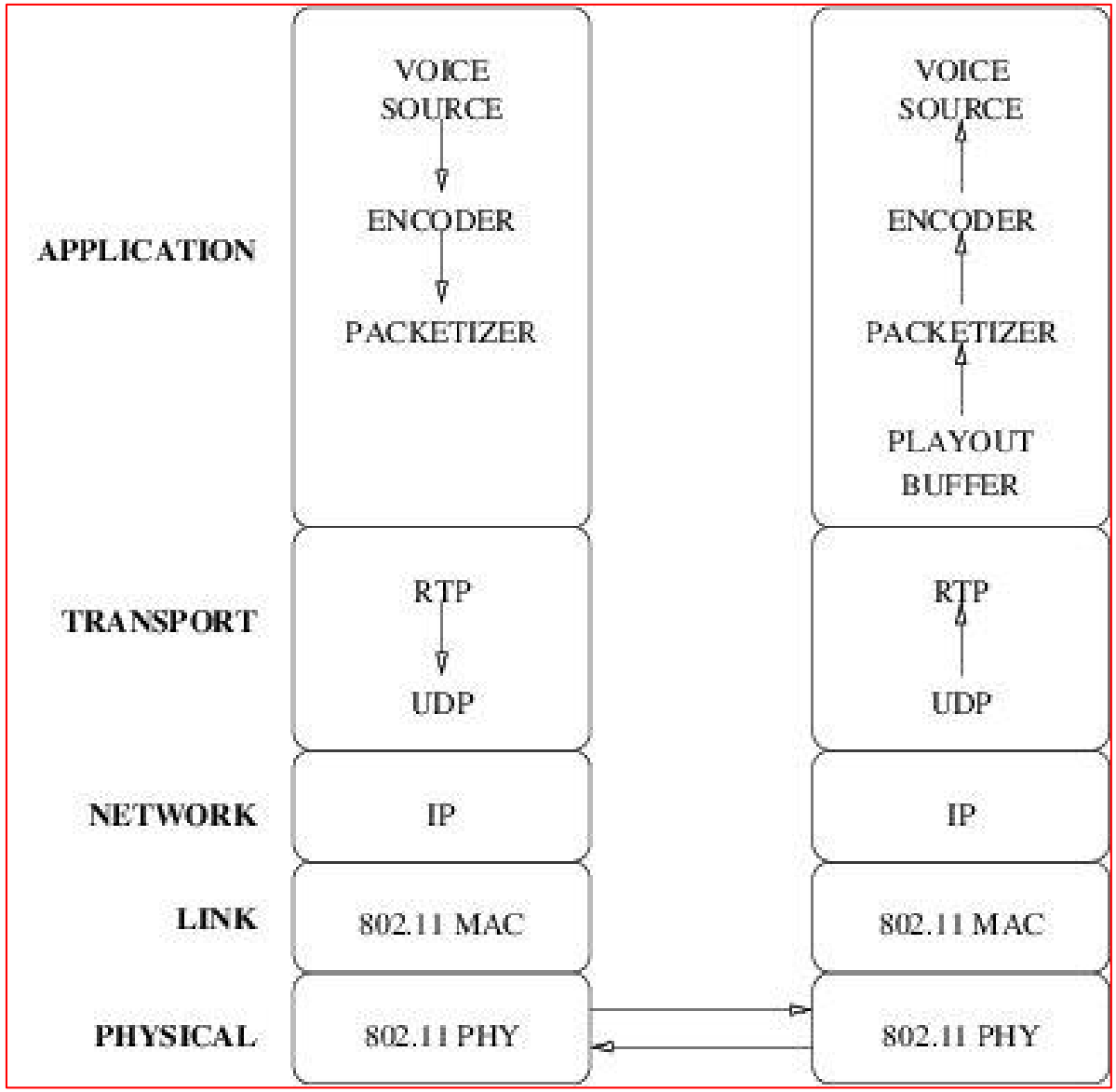
Telecommunication Systems
ICE 3109

Lecture 15
VoIP, Its Protocols and Components

VoIP Protocols



Complete protocol stack in implementing VoIP over IEEE 802.11

VoIP Protocols

Protocols Related to VoIP

	Audio/Video transport	Signaling
ITU-T	RTP/RTCP	H.323 H.248
IETF	RTP/RTCP	SIP MGCP MEGACO

VoIP Protocols

Audio/Video Transport Protocols: RTP and RTCP

- Real-time Transport Protocol (RTP) and Real Time Control Protocol (RTCP)
- Why RTP/RTCP??

✓ Jitter has to be taken into account by the receiver

✓ RTP is to compensate for jitter and de sequencing introduced by IP networks

✓ RTP includes:

- Information on the type of data transported (the 'payload').
- Timestamps
- Sequence numbers

	Audio/Video transport	Signaling
ITU-T	RTP/RTCP	H.323 H.248
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VoIP Protocols

Audio/Video Transport Protocols: RTP and RTCP

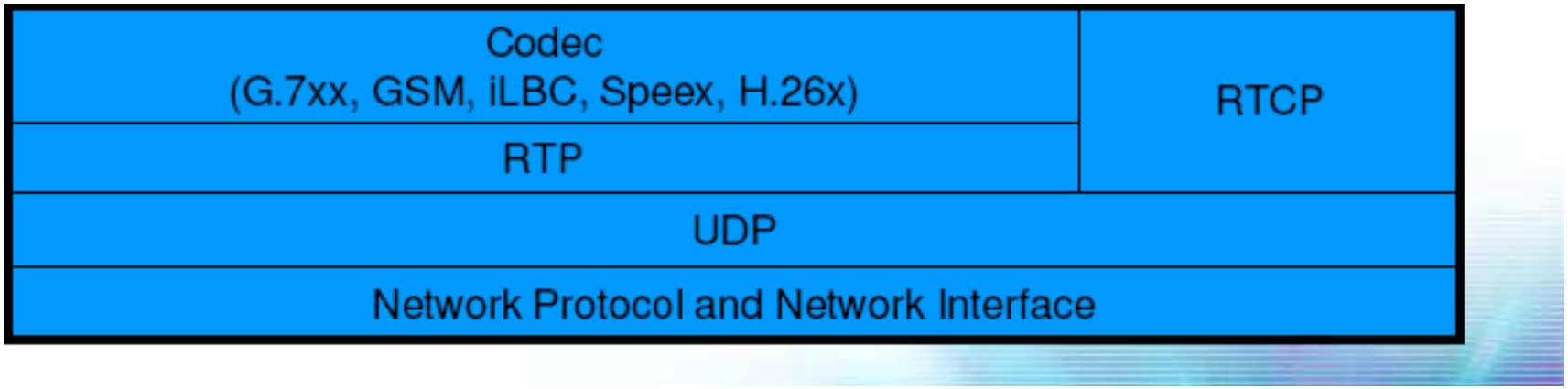
- RTCP carries some feedback on the quality of the transmission (the amount of jitter, the average packet loss, etc.) and some information on the identity of the participants as well
- RTP and RTCP do not have any influence on the behavior of the IP network and do not control quality of service in any way
- The network can drop, delay, or de sequence an RTP packet like any other IP packet

	Audio/Video transport	Signaling
ITU-T	RTP/RTCP	H.323 H.248
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VoIP Protocols

Audio/Video Transport Protocols: RTP and RTCP

- RTP/RTCP are always sent on top of UDP (User Datagram Protocol) on IP-based networks, not on the top of TCP



- TCP is for reliable data stream, for example, a file from a web server
- If your application doesn't care about corrupted or lost packets, and UDP is used to not incur the additional overhead
- VOIP is not significantly improved by reliable packet transmission
- Some cases things in TCP like retransmission and exponential back-off can actually hurt VOIP quality

VoIP Protocols

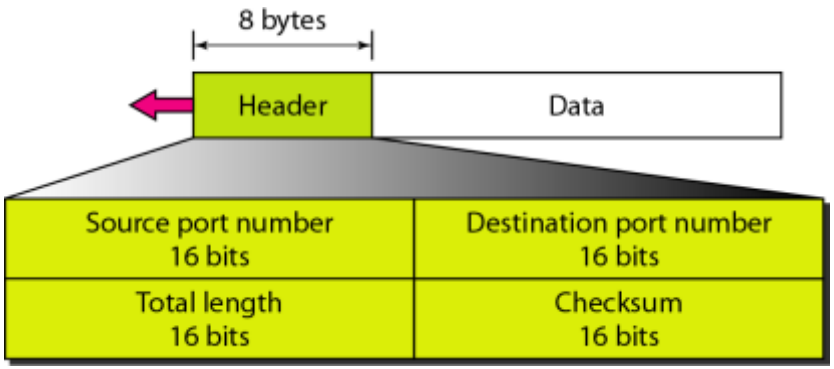
In a case where reliability is not of importance, UDP would make a good transport protocol - Why?

The User Datagram Protocol (UDP) is a connectionless, unreliable transport protocol. It performs very limited error checking.

If UDP is so powerless, why would a process want to use it?

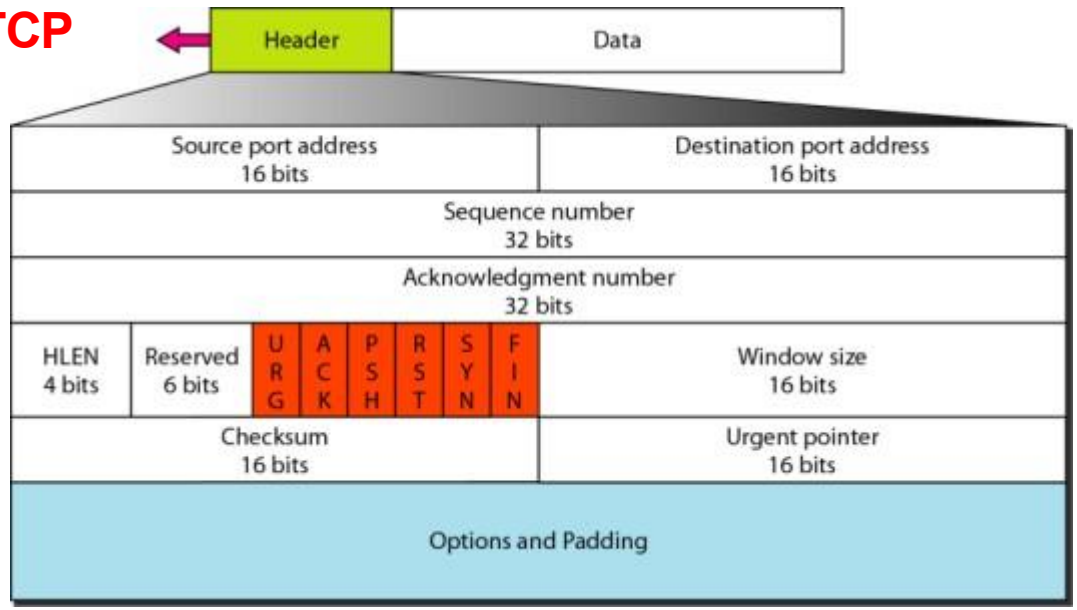
UDP is a very simple protocol using a minimum overhead. If a process wants to send a small message and does not care much about reliability, it can use UDP. Sending a small message by using UDP takes much less interaction between the sender and receiver than using TCP.

UDP



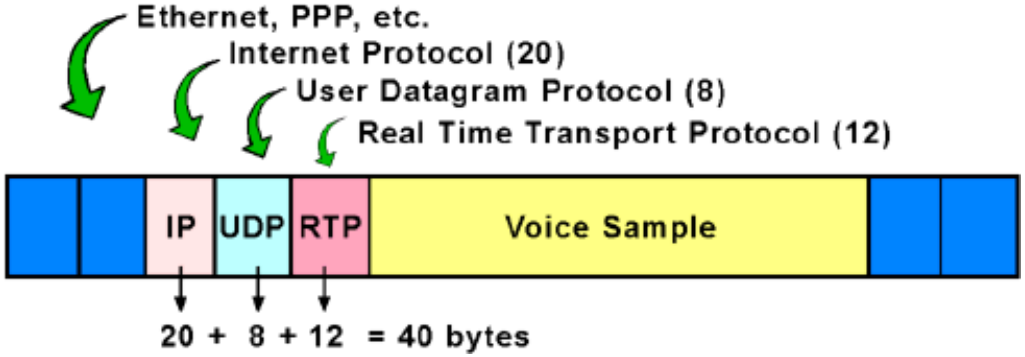
UDP header (8 bytes)

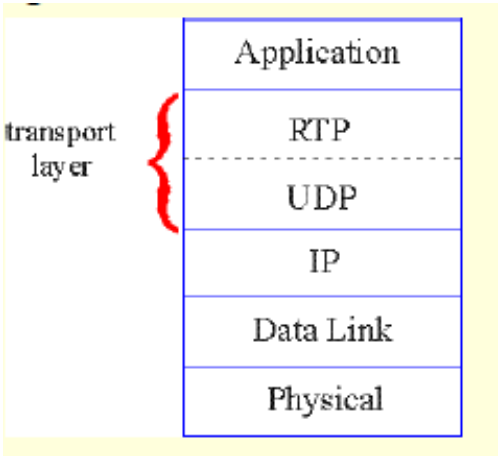
TCP



TCP header : 20 ~ 24 bytes

RTP (Real Time Transport Protocol)





RTP Header (12 bytes long)

Payload Type (7 bits)	Sequence Number (16 bits)	Time Stamp (32 bits)	Synchronization Source Identifier (SSRC) (32 bits)	Miscellaneous Fields (… bits)

- RTP defines a standardized packet format for delivering audio and video over IP networks
- RTP in its header adds additional information with the data packet to support some function, which does not incur significant delays in the packet transmission and reception
- RTP provides, **a) the payload format** (indicates the encoded format of the data), **b) sequence numbers** (for packet loss and reordering detection), **c) timestamps** (for synchronization), and **d) SSRC** (for identifying the source of the stream)

RTP

What is the need for RTP (Real-time Transport Protocol) in case of VoIP service?

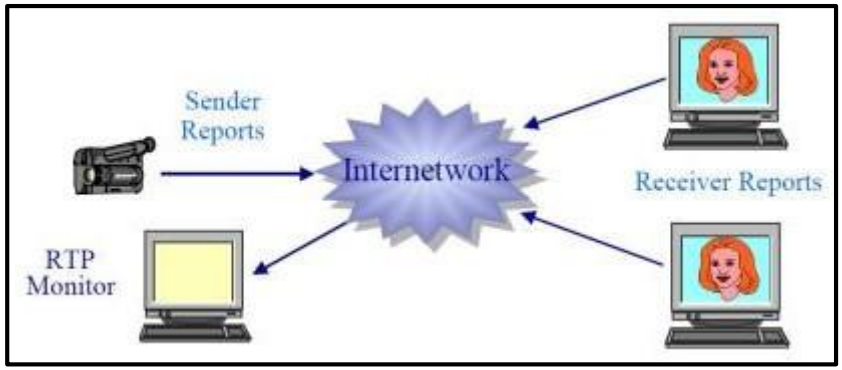
RTP Header (12 bytes long)

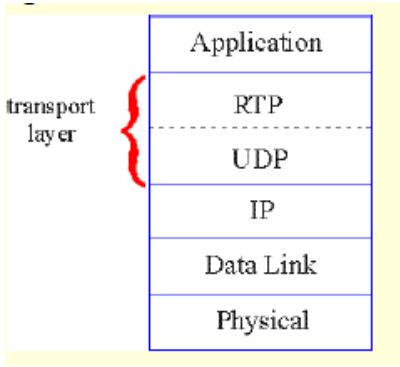
Payload Type (7 bits)	Sequence Number (16 bits)	Time Stamp (32 bits)	Synchronization Source Identifier (SSRC) (32 bits)	Miscellaneous Fields (… bits)

- Real-time multimedia streaming applications require timely delivery of information and can tolerate some packet loss
- Hence, it is required to have protocols which will format the data obtained from upper layer emphasizing on the timely delivery of data rather than the QoS and UDP is one of this type of protocols
- However, UDP guarantees no QoS. Therefore, RTP is used in conjunction with UDP to provide some sorts of QoS, such as, jitter compensation, detection of out of sequence arrival in data, synchronization, etc. At the same time, RTP does not impose any significant overhead. This is the main reason for using RTP.

RTCP

What is the purpose of RTCP?





- RTCP (Real-time control protocol) is a sister protocol (application layer) of the RTP
- RTCP is a tool to locate problems and evaluate where network performance is degrading
- Collects statistics on QoS of a session and feedback this data to the media source and other participants
- Such statistics include: transmitted octet and packet counts, lost packet counts, jitter, and round-trip delay time
- An application can use this information to control QoS parameters, perhaps by limiting flow, or using a different codec

VoIP Protocols

ITU-T Signaling Protocol: H.323 Protocol

- Defines how voice, video, and data traffic should be transported over IP-based LANs
- Addresses how delay-sensitive traffic such as voice and video get priority transport to ensure real-time communication service over the internet
- H.323 deals with three basic functional elements of VoIP

➢ Media gateway

➢ Media gateway controller (MGC) (in some settings this is called the *gatekeeper)*

➢ Signalling gateway

	Audio/Video transport	Signaling
ITU-T	RTP/RTCP	H.323 H.248
IETF	RTP/RTCP	SIP MGCP MEGACO

VoIP Protocols

ITU-T Signaling Protocol: H.323 Protocol

- H.323 is an umbrella protocol covering:
  - H.225 , which covers the setup of multimedia channels
  - H.245, which deals with the setup of single channel medium
  - RTP/RTCP
  - Audio/video codec

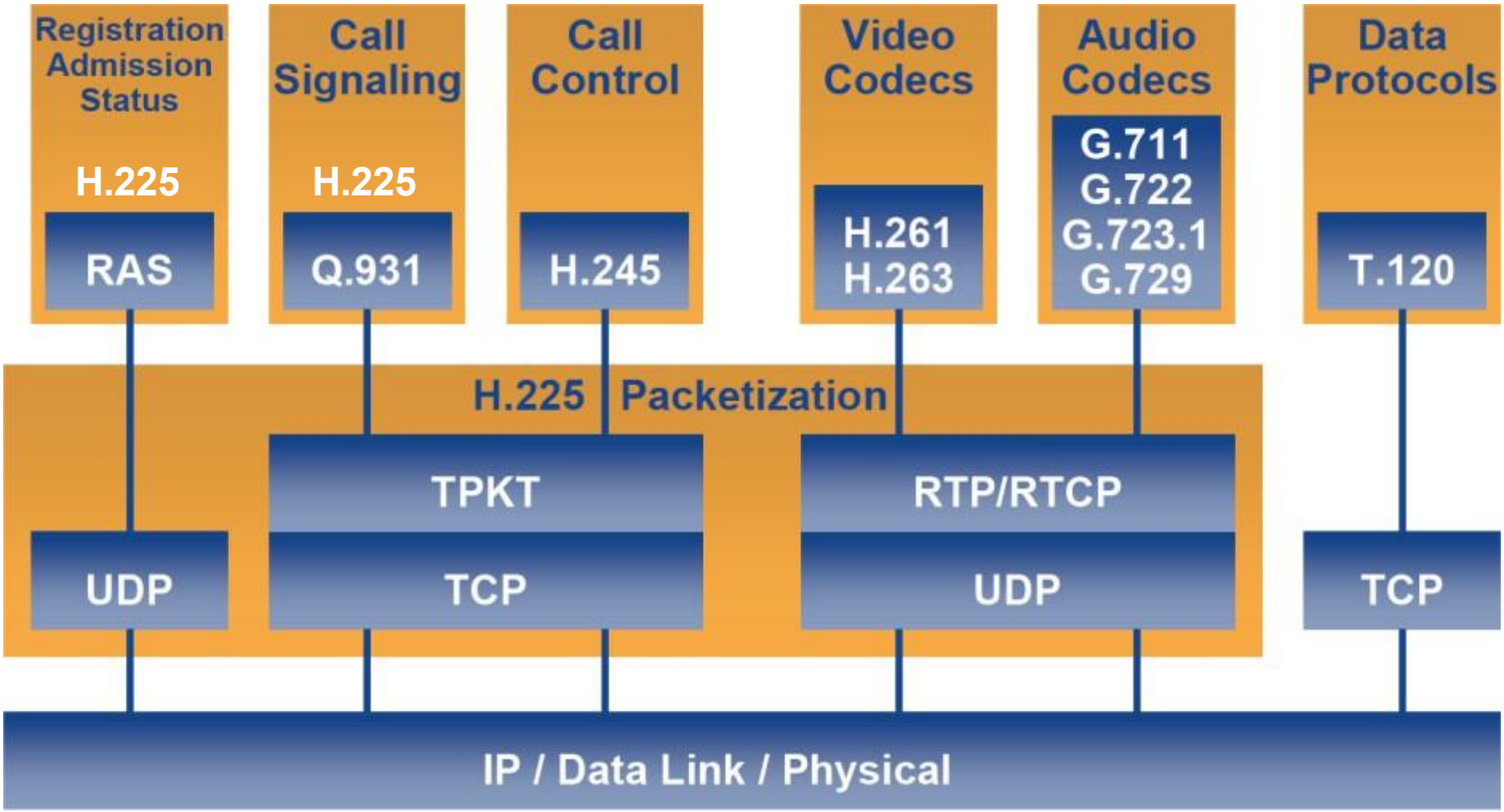
	Audio/Video transport	Signaling
ITU-T	RTP/RTCP	H.323 H.248
IETF	RTP/RTCP	SIP MGCP MEGACO

H.323 Protocol

- Defines how voice, video, and data traffic should be transported over IP-based LANs
- H.323 recommendation is based on RTP/RTCP for managing audio and video signals
- Addresses how delay-sensitive traffic, such as voice and video get priority transport to ensure real-time communication service over the Internet
- It also incorporates the ITU-T Rec. T.120 data-conferencing standard

	Audio/Video transport	Signaling
ITU-T	RTP/RTCP	H.323 H.248
IETF	RTP/RTCP	SIP MGCP MEGACO

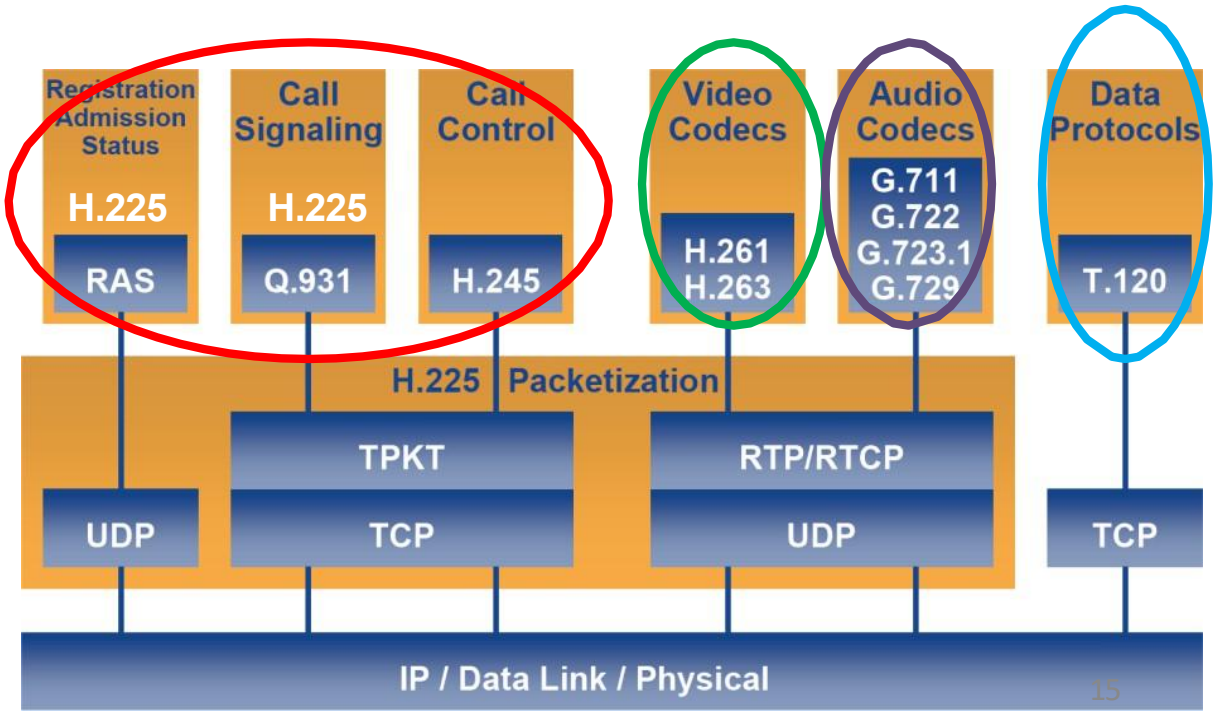
H.323 Protocol Suite



H.323 Protocol Suite

H.323 is an umbrella protocol covering:

- **H.225:** Uses ITU-T Q.931 protocol for call control signaling for multimedia channel setup between two H.323 devices
- **H.245:** Negotiates channel usage and capabilities (e.g., flow control, jitter management, preference requests) of the end devices
- **Audio codecs:** G.711, G.722, G.723.1, G.728, G.729
- **Video codecs:** H.261, H.263 (optional)
- **Text codecs (real-time):** T.120 suite (T.127, T.128, T.134, etc.)
- **RTP/RTCP**



VoIP Technologies

VoIP Technology

Many of the early IP technology products were initially based on proprietary protocols built on top of basic IP messaging. This quickly became a serious limiting factor and the counter to the core success factor of the Internet and its ability to connect to many through fairly simple open standards.

As an outgrowth of this need for standards the International Telecommunications Union (ITU) has introduced **H.323** as an umbrella standard for the multimedia communications over Local Area Network (LAN).

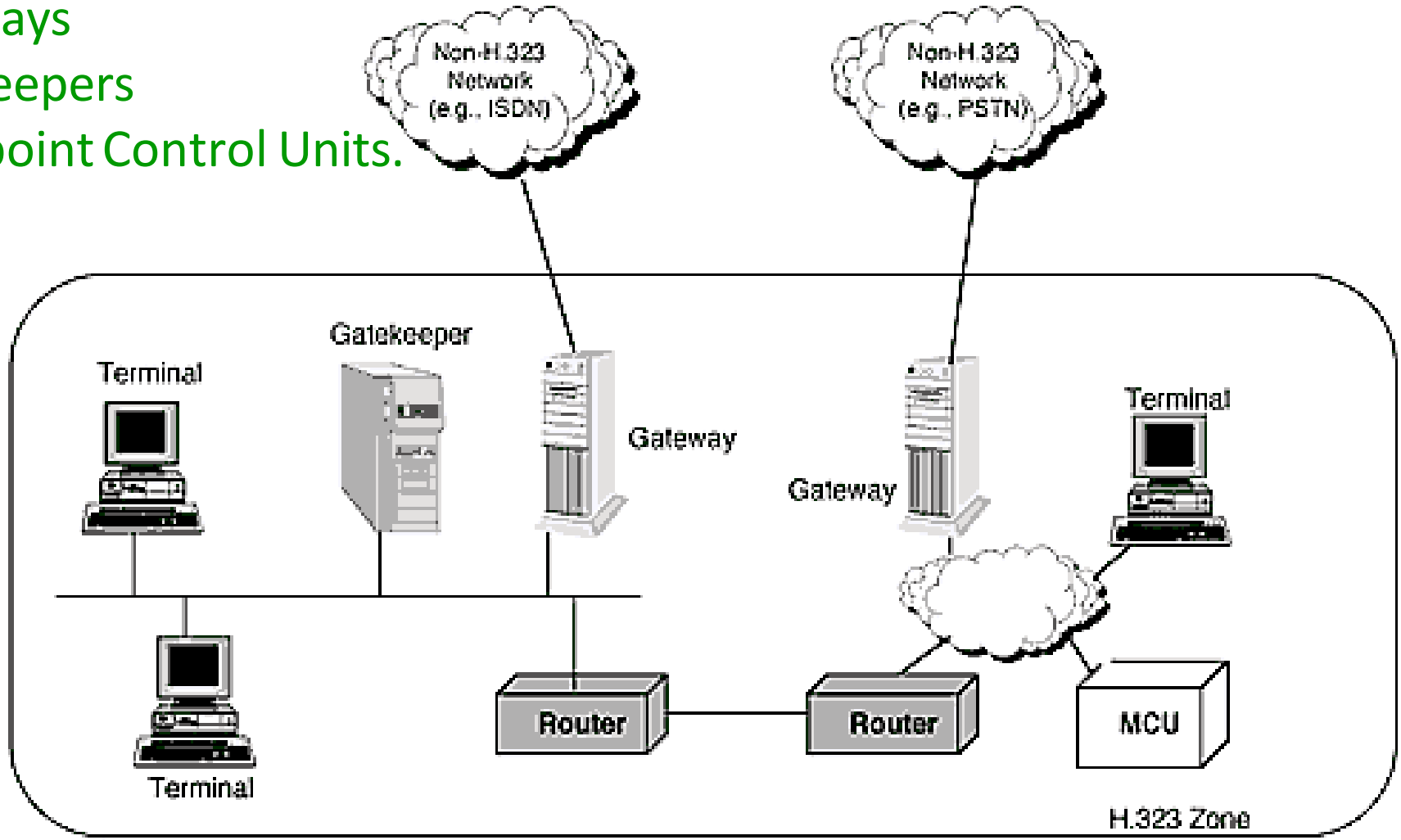
Architecture

The **H.323 standard specifies four** kinds of **components**, which when networked together, provide the point-to-point and point-to-multipoint multimedia-communication services:

H.323: Architectural Elements

H.323 deals with the following basic functional elements of VoIP:

- Terminals
- Gateways
- Gatekeepers
- Multipoint Control Units.



VoIP Technologies

Architecture (VoIP Components/Elements)

The function of each of the components/elements will be described next.

Terminals/End devices

These may be either traditional telephones (analog/GSM/ISDN/...), audio-equipped personal computers (PCs), or single use appliances.

Used for real-time bi-directional multimedia communications, an H.323 terminal can either be a PC or a stand-alone device, running an H.323 protocol stack and the multimedia applications.

VoIP Technologies

Architecture (VoIP Components/Elements)

An H.323 terminal can be an IP LAN Phone or a wireless IP device. Fig. 8.7 shows IP Phones that are used as Terminals in an H.323 Network.

The primary goal of H.323 is to interoperate with other multimedia terminals.



Fig. 8.7: IP Phones used as Terminals in an H.323 Network.

Gateways

If a traditional telephone is used at either calling side the call (i.e., its transmission format, signaling procedures, audio codecs) has to be translated to/from the format for transport over the Internet; these are the tasks of a gateway (GW).

VoIP Technologies

Architectural Elements

Fig. 8.8 shows the position of a gateway in between an IP network and a telephony network.

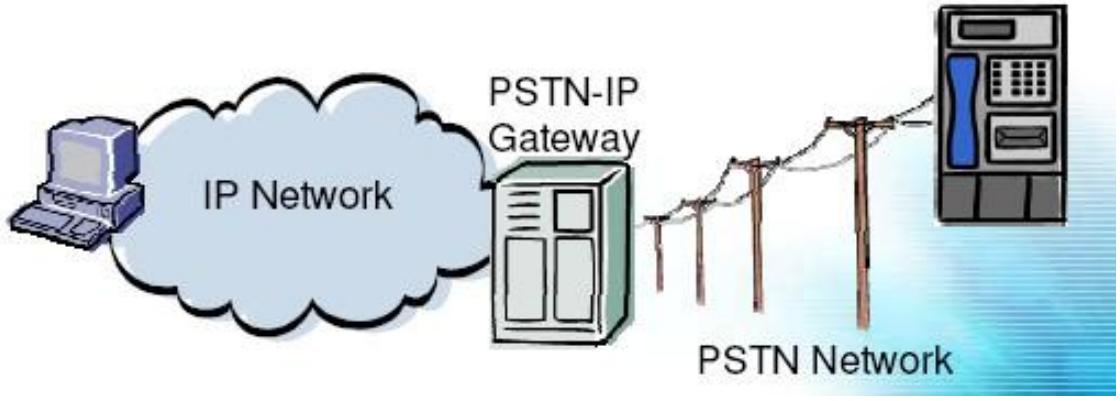


Fig. 8.8: A gateway in between an IP network and a PSTN.

A gateway connects two dissimilar networks. An H.323 gateway provides connectivity between an H.323 network and a non-H.323 network. For example, a gateway can connect and provide communication between an H.323 terminal and PSTN.

VoIP Technologies

Architectural Elements

This connectivity of dissimilar networks is achieved by translating protocols for call setup and release, converting media formats between different networks, and transferring information between the networks connected by the gateway.

A gateway is not required, however, for communication between two terminals on an H.323 network.

Gatekeepers/Proxies

The gatekeepers/proxies provide centralized call management functions; they may provide call admission control, bandwidth management, address translation, authentication, user location, etc.

VoIP Technologies

Architectural Elements

Gatekeepers/Proxies

Fig. 8.9 shows an H.323 gatekeeper in an H.323 network.

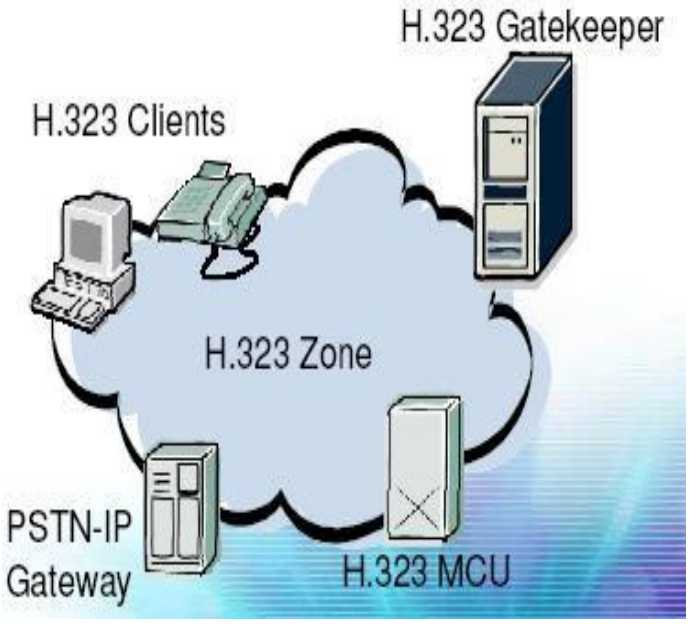


Fig. 8.9: An H.323 gatekeeper in an H.323 zone.

A gatekeeper can be considered the *"brain"* of the H.323 network. It is the focal point for all calls within H.323 network. Although they are not required, gatekeepers provide important services, such as addressing, authorization and authentication of terminals and gateways, bandwidth management, accounting, billing and charging.

VoIP Technologies

Architectural Elements

Gatekeepers/Proxies

Gatekeepers may also provide call-routing services. The gatekeeper is the cornerstone for the new IP switches. Similar to the feature lists switch vendors produce to differentiate their products today, the gatekeeper will be the primary gathering point for differentiating features and applications.

Multipoint Control Units (MCUs)

MCUs provide support for conferences of three or more H.323 terminals. All terminals participating in the conference establish a connection with the MCU.

The MCU manages conference resources, negotiates between terminals for the purpose of determining the audio or video codec to use, and may handle the media stream.

VoIP Technologies

Architectural Elements

Multipoint Control Units (MCUs)

The position of the MCU is shown in Fig. 8.10.

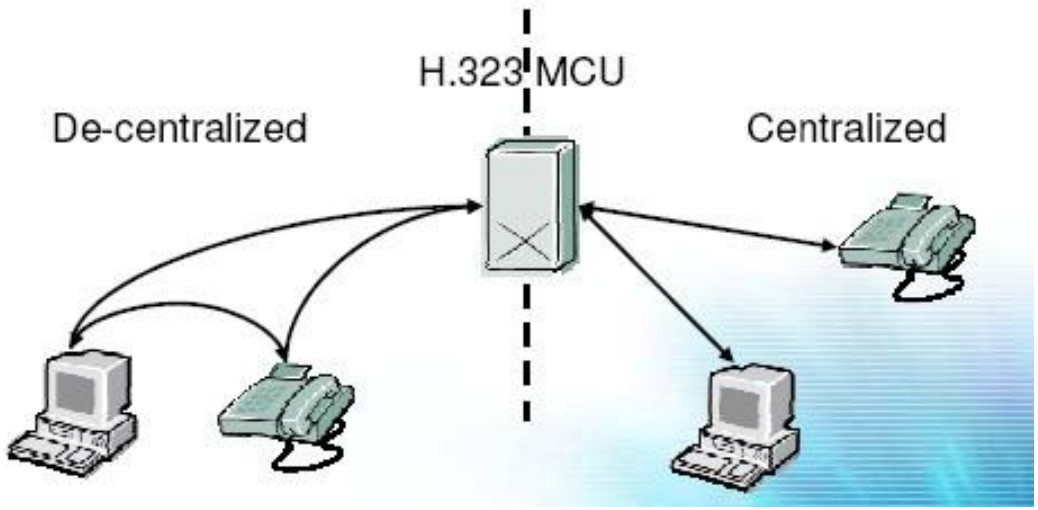


Fig. 8.10: An H.323 MCU connecting different H.323 clients for multipoint conferencing.

The gatekeepers, gateways, and MCUs are logically separate components of the H.323 standard but can be implemented as a single physical device. The components may be implemented as hardware or software and may be integrated into single units optionally.

VoIP Technologies

VoIP Technology

They communicate with each other over signaling and voice-transporting **protocols**. To ensure interoperability between products of different vendors, standardization bodies have elaborated standards for both classes of protocols.

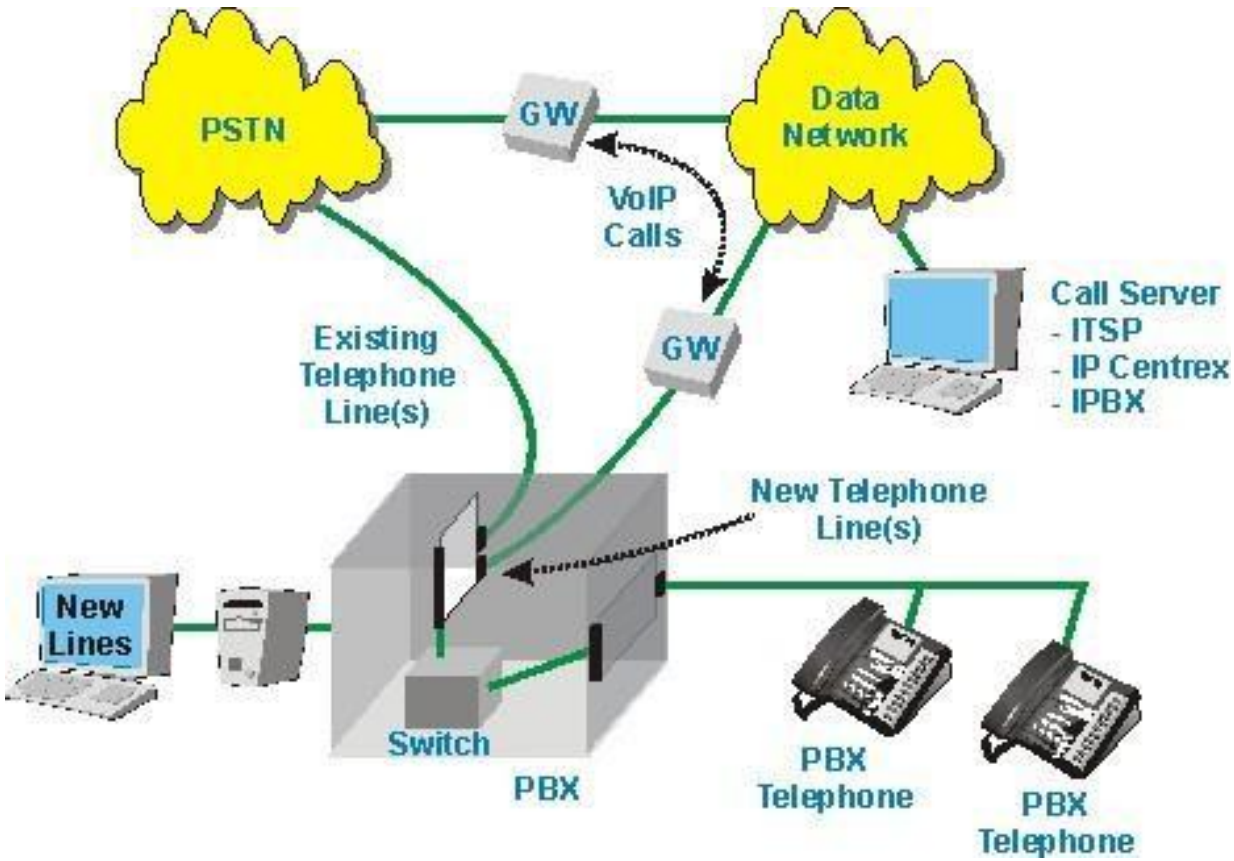


Fig. 8.11: The interconnection of IP WAN with PSTN and VPN.

VoIP Technologies

VoIP Technology

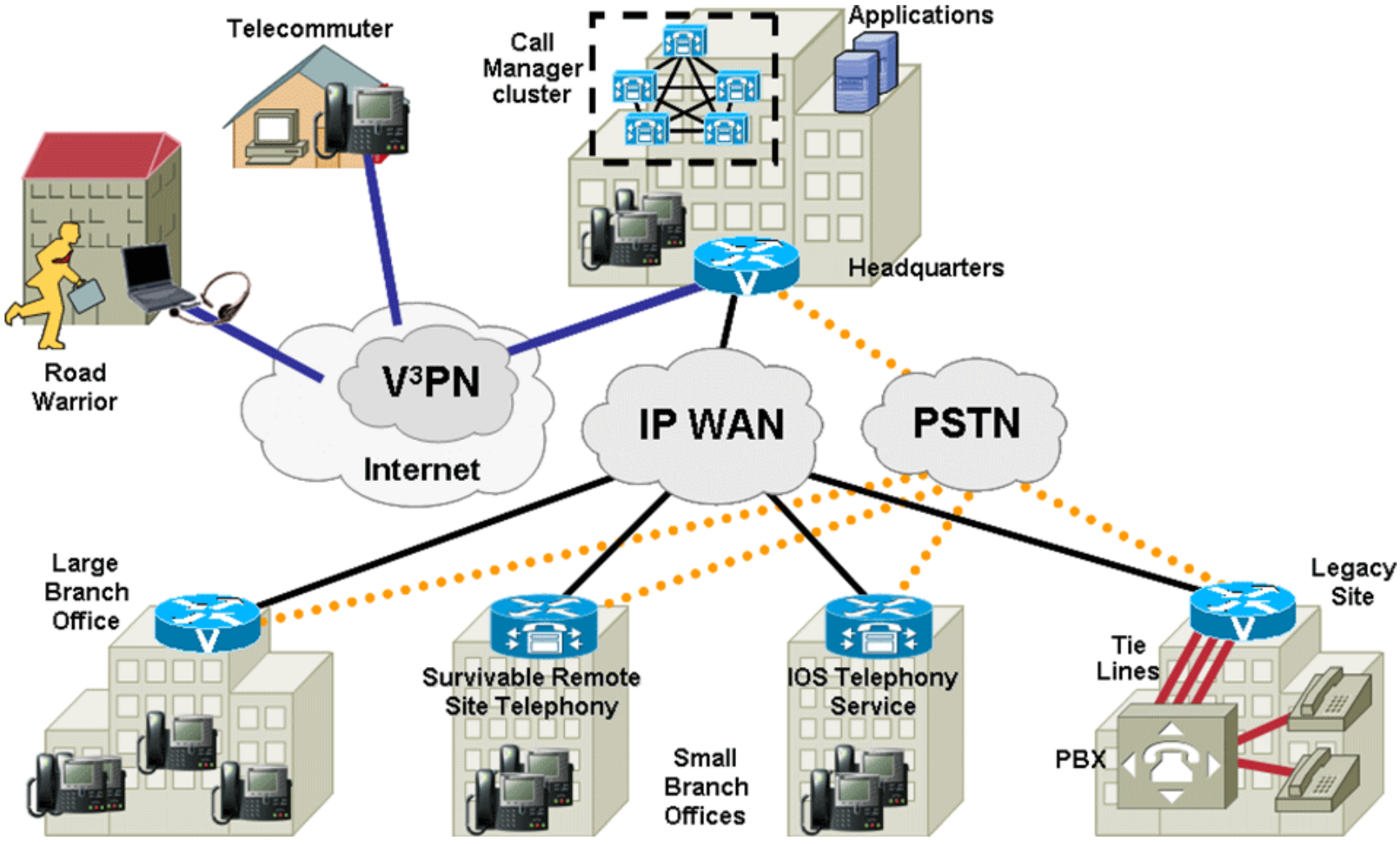
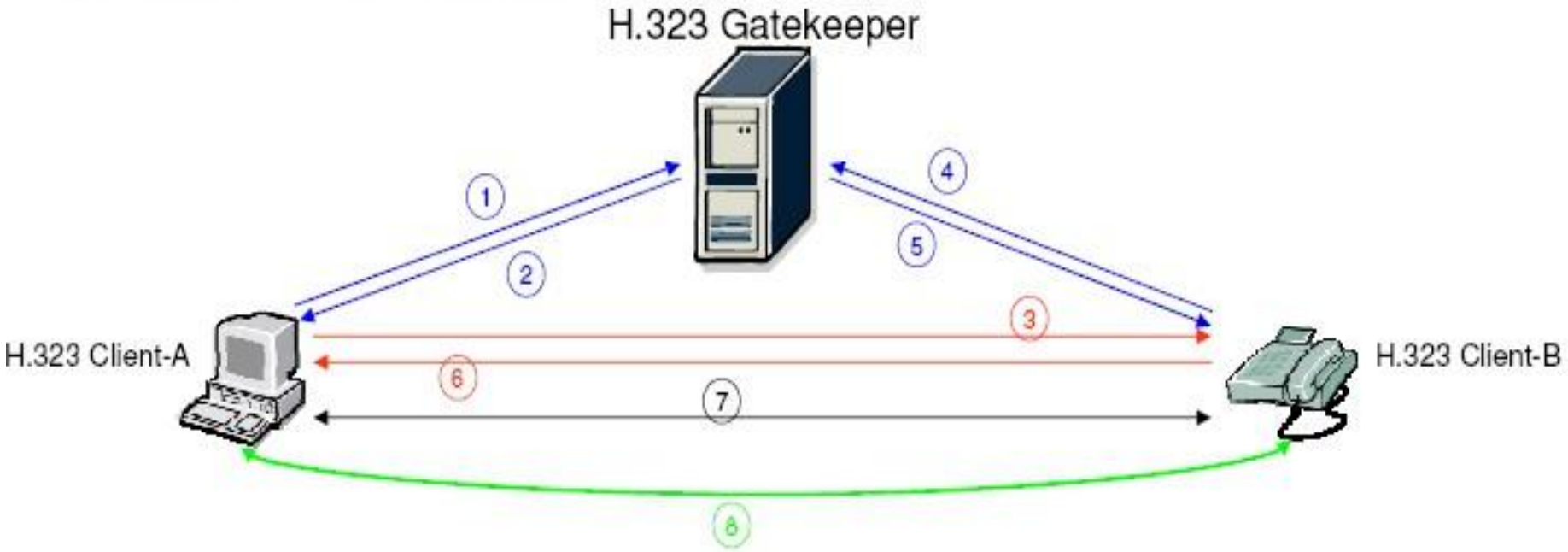


Fig. 8.12: Interworking of IP Telephony with VPN and PSTN.

VoIP Protocols

H.323 Example

1. Client-A sends ARQ (Admission ReQuest) to Gatekeeper: it asks to be connected to Client-B alias
2. Gatekeeper confirms or rejects the call (ACF, Admission ConFirm / ARJ, Admission ReJect): it returns the Client-B call signaling address (IP address and port of the call signaling address)
3. Client-A sends a SETUP to Client-B call signaling Address



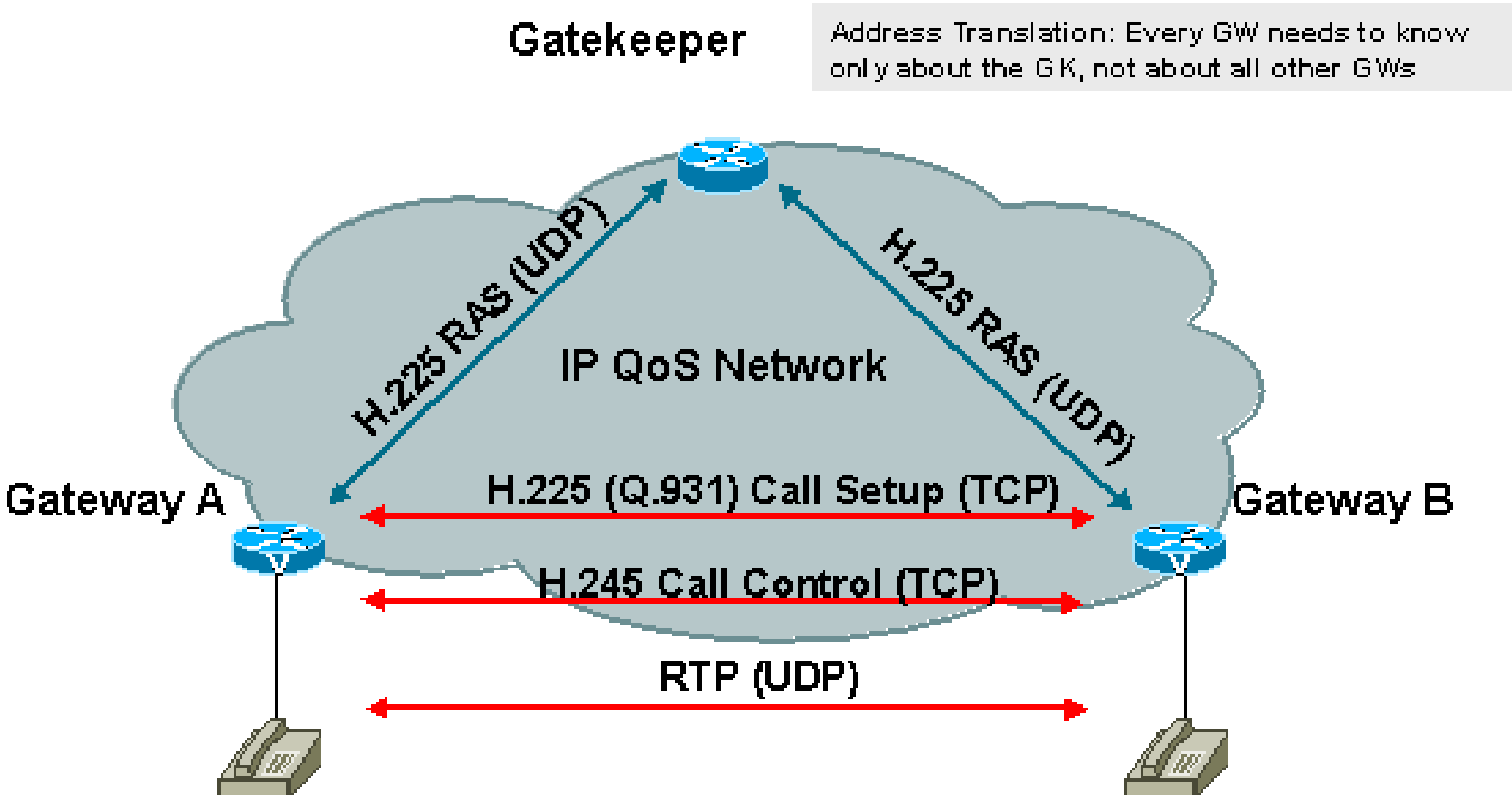
Client B

4. Client-A sends ARQ (Admission ReQuest) to Gatekeeper: it asks to be connected to Client-A (he knows already Alias and call signaling address)
5. Gatekeeper confirms or rejects the call (ACF, Admission ConFirm / ARJ, Admission ReJect): it returns the Client-A call signaling address (IP address and port) if it was asked
6. Client-B sends a CONNECT to Client-A call signaling address
7. Capabilities about the call are exchanged among the terminals (on H.245 channel address different from call signaling channel address)
8. The media is sent end-to-end; i.e. to the IP addresses and ports negotiated during the signaling

Messages 3, 6, 7 may be exchanged with the H.323 Gatekeeper depending on the call model adopted
(Direct signaling, Gatekeeper-routed call signaling, Gatekeeper-routed H.245 control)

VoIP Protocols

H.323 Example



VoIP Protocols

IETF-Signaling Protocol: SIP and MGCP

A. Session Initiation Protocol (SIP)

B. Media Gateway Control Protocol (MGCP)

A. Session Initiation Protocol (SIP)

- Application layer signalling protocol
- Deals with interactive multimedia communication sessions between end-users, called *user agents*
- Defines initiation, modification, and termination of sessions
- Basic scope of SIP is to exchange IP addresses and port numbers to which systems can receive data.

VoIP Protocols

A. Session Initiation Protocol (SIP)

SIP is based on RFC 2543 and is an application layer signaling protocol. It deals with interactive multimedia communication sessions between end-users, called *user agents*. It defines their initiation, modification, and termination. SIP calls may be terminal-to-terminal, or they may require a server to intercede.

If a server is to be involved, it is only required to locate the called party. For interworking with non-IP networks, Megaco and H.323 are required. Often vendors of VoIP equipment integrate all three protocols on a single platform.

VoIP Protocols

SIP

SIP is closely related to IP. SIP borrows most of its syntax and semantics from the familiar HTTP (hypertext transfer protocol). An SIP message looks very much like an HTTP message, especially with message formatting, header, and multipurpose Internet mail extension support. It uses addresses that are very similar to URLs (uniform resource locators) and to email. For example, a call may be made to so-in-so@such-and-such. SIP messages are text-based rather than binary. This makes writing easier and the debugging of software more straightforward.

VoIP Protocols

SIP

There are two modes with which a caller can setup a call with SIP. These are called *redirect* and *proxy*, and servers are designed to handle these modes. Both modes issue an "invite" message for another user to participate in a call. The redirect server is used to supply the address (URL) of an unknown called addressee.

In this case the "invite" message is sent to the redirect server, which consults the location server for address information. Once this address information is sent to the calling user, a second "invite" message is issued, now with the correct address.

One specific type of SIP is called SIP-T (T for telephone). This is a function that allows calls from CCITT Signaling System 7 (SS7) to interface with telephone in an IP-based network. The particular user part of SS7 for this application is ISUP.

VoIP Protocols

SIP Architectural Elements

- **User Agent Client (UAC)**

✓ A logical entity that creates a new request, and then uses the client transaction state machinery to send it

- **User Agent Server (UAS)**

✓ A logical entity that generates a response to a SIP request

✓ The response accepts, rejects, or redirects the request

- **User Agent (UA)**

✓ A logical entity that can act as both a user agent client and user agent server (UA = UAC + UAS)

- **Proxy Servers**

✓ A logical entity that routes SIP requests to UASs and SIP responses to UACs. It primarily plays the role of routing, i.e., to ensure that a request is sent to another entity "closer" to the targeted user. A proxy interprets, and, if necessary, rewrites specific parts of a request message before forwarding it.

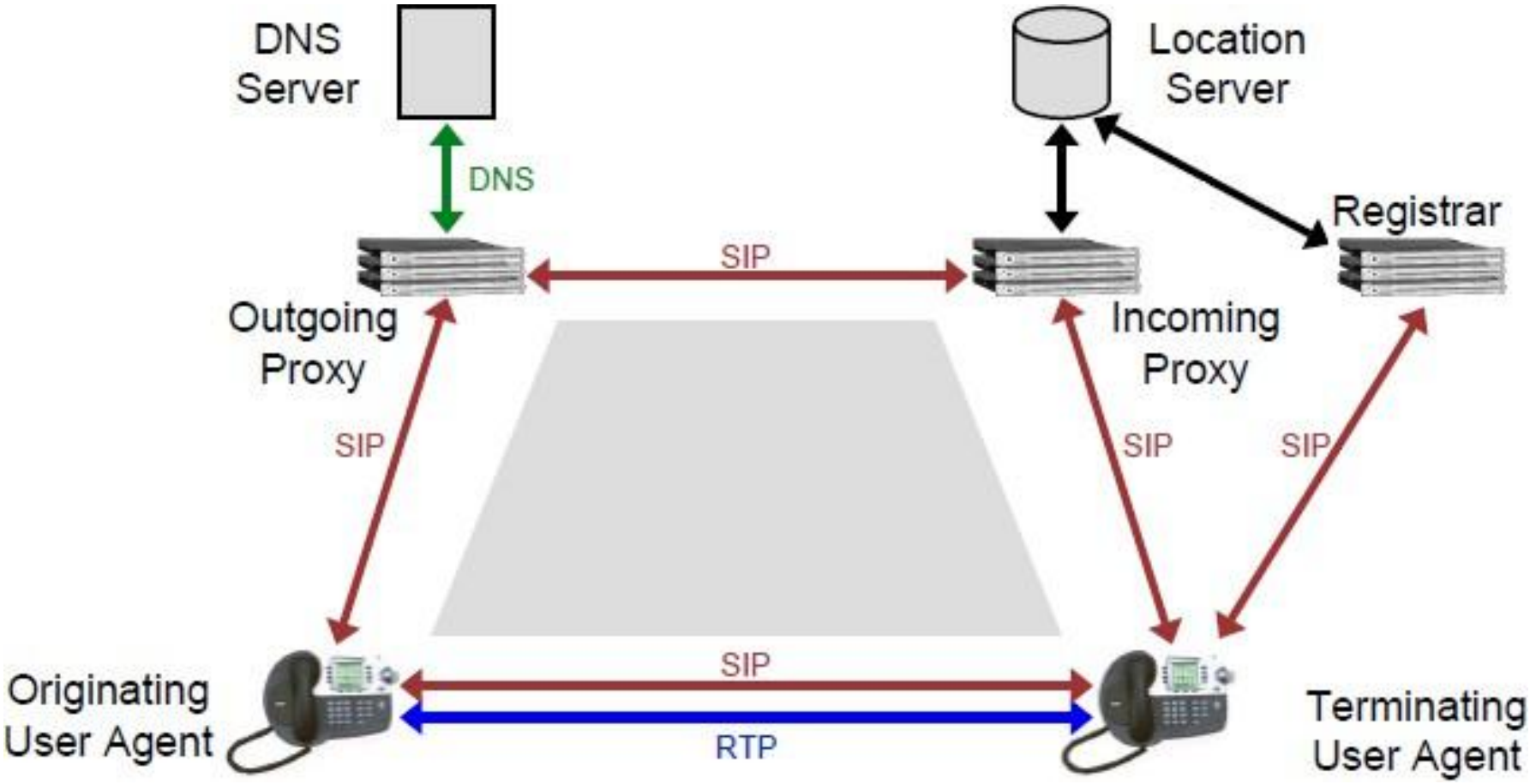
- **Register Servers**

✓ Accepts REGISTER requests and places the information it receives in those requests into the location service for the domain it handles

- **Location Server**

✓ It records and maintains the contact information of each and every UA within a typical enterprise and helps in providing the exact location of UA.

SIP: Architectural Elements



SIP (Session Initiation Protocol

- Application layer IETF's signalling protocol
- Deals with interactive multimedia communication sessions between end-users, called *user agents*
- Defines initiation, modification and termination of real-time sessions
- SIP calls may be terminal-to-terminal, or they may require a server to intercede (mediate). If a server is to be involved, it is only required to locate the called party. For interworking with non-IP networks, Megaco and H.323 are required.

VoIP Protocols

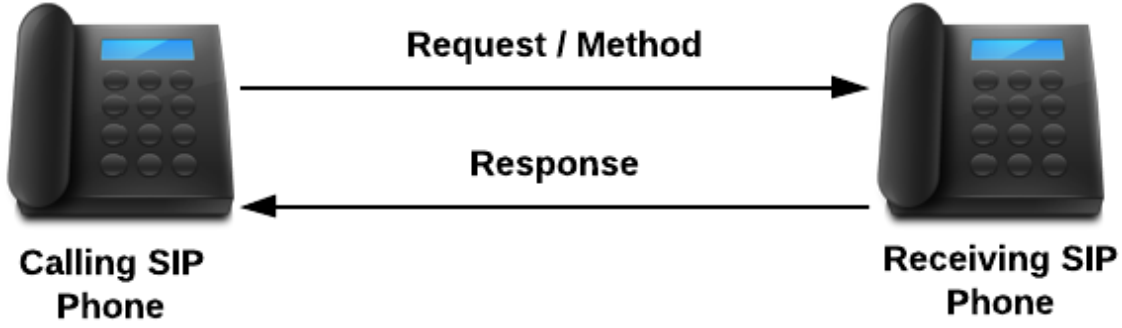
SIP Method

SIP makes uses of different types of messages (methods) to communicate among parties.

- **INVITE:** initiate sessions
- **ACK:** confirms session establishment
- **BYE:** terminates sessions
- **CANCEL:** cancels a pending INVITE
- **REGISTER:** binds a permanent address to a current location
- **OPTIONS:** capability inquiry.

SIP Method

- SIP makes uses of different types of messages (methods) to communicate among parties
  - ✓ **INVITE:** Initiates sessions by inviting another UA to a session
  - ✓ **ACK:** These requests correspond to an INVITE request. It is the final confirmation from the end system for session establishment.
  - ✓ **BYE:** It is used by calling and called parties to release (terminate) a call (session).
  - ✓ **CANCEL:** It enables to cancel any in-progress request.
  - ✓ **REGISTER:** It registers (binds) a users' current location with a SIP registrar server by sending user's address
  - ✓ **OPTIONS:** It enables to query and collect user agents and network server capabilities.



VoIP Protocols

IETF- Signaling Protocol: MGCP

B. Media Gateway Control Protocol (MGCP)

This protocol was the predecessor to Megaco (ITU-T Rec. H.248) and still holds sway with a number of carriers and other VoIP users. MGCP assumes a call control architecture where the call control "intelligence" is outside the gateways (i.e., at the network edge) and handled by external call control elements.

Thus, the MGCP assumes that these call control elements, or "call agents," will synchronize with each other to send coherent commands to the gateways under their command. There is no mechanism defined in MGCP for synchronizing "call agents." It is, in essence, a master/slave protocol where the gateways are expected to execute commands sent by the "call agents."

VoIP Protocols

MGCP

In the MGCP protocol an assumption is made that the connection model consists of constructs that are basic endpoints and connections. Endpoints are sources or sinks of data and could be physical or virtual. The following are **two examples** of **endpoints**:

1. An **interface on a gateway** that **terminates a trunk connected to a PSTN switch** (e.g., local or toll-connecting, etc.). A gateway that terminates trunks is called a ***trunk gateway***.

2. An **interface on a gateway** that **terminates an analog POTS** (plain old telephone service) connected to a telephone, a key system, PABX, and so on. A gateway that terminates residential POTS lines (to telephones) is called a *residential gateway*.

An **example** of a **virtual endpoint** is an audio source in an audio-content server. Creation of physical endpoints requires a hardware installation, while creation of virtual endpoints can be done in software.